

(1) $K_f=0$ 时, 系统的开环传递函数 $G(s) = \frac{10}{s(s+2)} = \frac{\omega_n^2}{s(s+2\xi\omega_n)}$, 可得:

$$\begin{cases} \omega_n = \sqrt{10} \\ \xi = \frac{\sqrt{10}}{10} \end{cases}$$

单位斜坡输入信号的稳态误差 e_{ss} :

$$e_{ss} = \lim_{s \rightarrow 0} sE(s) = \lim_{s \rightarrow 0} \frac{sR(s)}{1 + G(s)H(s)} = \lim_{s \rightarrow 0} \frac{s+2}{s^2 + 2s + 10} = 0.2$$

(2) $K_f \neq 0$ 时, 系统的开环传递函数 $G(s) = K_A \frac{\frac{1}{s(s+2)}}{1 + \frac{K_f s}{s(s+2)}} = \frac{K_A}{s(s+2+K_f)} = \frac{\omega_n^2}{s(s+2\xi\omega_n)}$, 可得:

$$\begin{cases} \omega_n = \sqrt{10} \\ K_f = \frac{6\sqrt{10}}{5} - 2 \end{cases}$$

$$e_{ss} = \lim_{s \rightarrow 0} sE(s) = \lim_{s \rightarrow 0} \frac{sR(s)}{1 + G(s)H(s)} = \lim_{s \rightarrow 0} \frac{s + \frac{6\sqrt{10}}{5}}{s^2 + \frac{6\sqrt{10}}{5}s + 10} = \frac{3\sqrt{10}}{25}$$

(3) 系统的开环传递函数 $G(s) = \frac{\omega_n^2}{s(s+2\xi\omega_n)} = \frac{K_A}{s(s+2+K_f)}$, 稳态误差 $e_{ss} =$

$$\lim_{s \rightarrow 0} sE(s) = \frac{2+K_f}{K_A} = 0.2, \text{ 可得:}$$

$$\begin{cases} K_A = 36 \\ K_f = 5.2 \end{cases}$$